School Inputs

Both administrative and survey data show considerable improvements in most input-based measures of schooling quality. But there is very little impact of these improvements in school facilities on learning outcomes. This is not to suggest that school infrastructure does not matter for improving learning outcomes (they may be necessary but not sufficient), but the results highlight that infrastructure by itself is unlikely to have a significant impact on improving learning levels and trajectories. Similarly, while there may be good social and humanitarian reasons for 'mid-day meal' programmes (including nutrition and child welfare), there is no evidence to suggest that they improve learning outcomes. Even more striking is the fact that no credible study on education in India has found any significant positive relationship between teachers possessing formal teacher training credentials and their effectiveness for improving student learning.

In the same way, there is no correlation between teacher salary and its effectiveness for improving student learning, and at best there are very modest positive effects of reducing pupil-teacher ratios on learning outcomes. As discussed further, these very stark findings most likely reflect *weaknesses in pedagogy* and *governance* which are key barriers in translating increased spending into better outcomes.

The results summarised so far can be quite discouraging. Fortunately, the news is not all bad, because the evidence of the past decade also points consistently to interventions that have been highly effective for improving learning outcomes, and are able to do so in much more cost-effective ways than the status-quo patterns of spending.

Pedagogy

The science of education, teaching and classroom instruction (pedagogy) is a key determinant which decides how schooling inputs translate into learning outcomes. Today, following the right kind of pedagogy has become particularly challenging in India as several millions of first-generation learners have joined a rapidly expanding national schooling system. In particular, standard curricula, textbooks, and teaching practices that may